Project Proposal

Data Visualization

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Basic Information

Project Title - Dominance Redefined: Visualizing Max Verstappen's 2023 Formula 1 Season

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GitHub Repo link - https://varun1421.github.io/

Background & Motivation

For this project, I intend to analyze Max Verstappen's 2023 Formula 1 season through data visualization. I selected this topic due to my personal interest in Formula 1 and Verstappen is my favorite driver. Beyond personal interest, the 2023 season presents a compelling case for visualization as it provides a clear narrative of dominance, consistency, and competitive separation throughout a single championship year.

This topic is particularly appealing from a visualization perspective because Formula 1 data inherently supports various forms of analysis: race-by-race progression, comparisons with other drivers, cumulative point accumulation, finishing consistency, and qualifying versus race outcomes. Verstappen's 2023 season is especially suitable because it enables me to examine not only his frequency of victories but also the evolution of his performance over time and its comparison with the rest of the grid.

My motivation is to transform a sports season that fans recall as "dominant" into a measurable and explorative data set. Rather than merely asserting Verstappen's dominance, I aim to demonstrate how that dominance manifested in the data: through finishing positions, cumulative points, consistency across races, and the widening disparity between him and his competitors. Additionally, I seek to provide users with interactive capabilities to comprehend the location of his advantage's strength and whether that dominance was consistent or concentrated in specific calendar periods.

Project Objectives

Objective 1 - Analyze Max Verstappen's race-by-race performance across the 2023 Formula 1 season using finishing position, points, and podium outcomes.

Objective 2: Compare Verstappen's performance with other top drivers in the 2023 season to understand the scale of his dominance.

Objective 3: Examine how Verstappen's lead developed over time by visualizing cumulative points and changes in the championship gap throughout the season.

Objective 4: Show the relationship between qualifying performance and race outcomes in order to understand whether Verstappen's dominance came mainly from strong starts, race pace, or overall consistency.

Data

The project will utilize publicly available Formula 1 data for the 2023 season sourced from Kaggle. The primary dataset used is:

- 2023 F1 Season (JDAustralia)

Dataset Description

This dataset contains comprehensive information about the 2023 Formula 1 season, including:

- Race results and finishing positions
- Driver standings and cumulative points
- Sprint race points (provided in separate files)

The dataset is organized across multiple CSV files, which separate different aspects of the season such as race results and sprint points. This structure allows for flexible data integration and analysis.

How the Dataset Will Be Used

The dataset will be used to:

- Extract race-by-race results for Max Verstappen and other drivers
- Track cumulative points progression across the season
- Compare Verstappen's performance with other top drivers
- Analyze finishing consistency and race outcomes

Derived metrics such as cumulative points, points gaps between drivers, and qualifying-to-finish differences will be computed during data processing.

Relevance to the Project

This dataset is appropriate because it provides a structured and complete view of the 2023 season, enabling both race-level analysis and longitudinal tracking of driver performance. It supports all project objectives, particularly the analysis of dominance through cumulative performance and comparison with other drivers.

Data Processing

The dataset used for this project is already well-structured, so the amount of data cleaning required is expected to be **minimal**. The CSV files provide clearly organized information about race results, standings, and sprint points, and most variables are already in a usable format.

Data Cleaning

Since the dataset is relatively clean, only light preprocessing will be required:

- Ensure that all data corresponds to the **2023 Formula 1 season**
- Verify that driver names and race identifiers are consistent across files
- Sort races correctly by round to maintain proper chronological order
- Remove any unnecessary columns that are not relevant to the analysis

Because the dataset already organizes drivers based on points and standings, no major restructuring or correction is expected.

Derived Quantities

Only a small number of additional variables will be created if needed. The main derived quantity will be:

- Total points at the end of the season for each driver

Additional simple metrics such as cumulative points or race-by-race comparisons may also be computed if required for specific visualizations, but most necessary values are already present in the dataset.

Tools, Languages, and Techniques

The dataset will be processed using:

- **VS Code** along with relevant extensions for editing and data handling
- Basic data manipulation techniques to verify, organize, and prepare the dataset
- Optional use of lightweight tools or scripts if additional calculations are required

Visualization of data

Proceed to the last page

Must have Features

Cumulative Points Progression

This feature will display a line chart showing cumulative points over the season for Max Verstappen and selected top drivers.

- Supports Objective: Objective 3
- **Purpose:** This allows users to see how Verstappen's lead developed over time and identify when he separated from the rest of the field.

Comparative Driver Summary

This feature will compare Verstappen with other top drivers using summary statistics such as:

- Total points
- Number of wins
- Number of podium finishes
- Average finishing position
- Supports Objective: Objective 2
- **Purpose:** It provides a clear and direct way to quantify how dominant Verstappen was compared to his competitors.

Qualifying vs Finishing Analysis

This feature will visualize the relationship between qualifying position and finishing position. I believe this will more or less check positions gained/lost.

- Supports Objective: Objective 4
- **Purpose:**

  It helps determine whether Verstappen’s success was driven mainly by strong qualifying or by race-day performance and consistency.

Optional Features

Tooltips with Detailed Race Information

Hover interactions will display:

- Race name
- Driver
- Finishing position
- Points scored

I feel like this enhances user experience without cluttering the visualization.

Focus + Context Timeline

A zoomable timeline allowing users to:

- Explore specific parts of the season
- Maintain awareness of the full season

Useful for long time-series data.

Project Schedule

Week 1 (Apr 3 – Apr 10) – Proposal Phase

Deadline: Revised Proposal – Apr 10

- Finalize project topic (Verstappen 2023 analysis)
- Review and understand dataset (JDAustralia Kaggle dataset)
- Complete all proposal sections:
    - Objectives
    - Data
    - Data Processing
    - Visualization Design
- Develop initial sketches using the Five Design Sheet methodology
- Submit revised proposal

Week 2 (Apr 11 – Apr 17) – Initial Development

Deadline: Alpha Release – Apr 17

- Load dataset into VS Code
- Perform initial data verification and organization
- Compute key derived values (cumulative points, finishing positions, etc.)
- Create basic draft visualizations (line charts, bump charts)
- Begin implementing core components of the dashboard
- Submit Alpha Release with initial working visualizations

Week 3 (Apr 18 – Apr 24) – Core Feature Development

- Refine and finalize data structure

- Implement main visualizations:
    - Cumulative points progression
    - Race-by-race performance
- Begin implementing comparative driver analysis
- Improve layout and structure of visualization

Week 4 (Apr 25 – Apr 30) – Integration & Interaction

Deadline: Beta Release – Apr 30

- Add interactivity:
    - Driver selection
    - Highlighting and filtering
- Implement qualifying vs finishing analysis
- Integrate all visual components into one dashboard
- Improve design (colors, labels, readability)
- Submit Beta Release

Week 5 (May 1 – May 7) – Refinement & Polishing

- Debug and refine visualizations
- Improve usability and interaction flow
- Add optional features:
    - Tooltips
    - Annotations
    - Summary statistics
- Test for consistency and clarity

Week 6 (May 8 – May 14) – Presentation Preparation

Deadline: Project Presentation – May 11

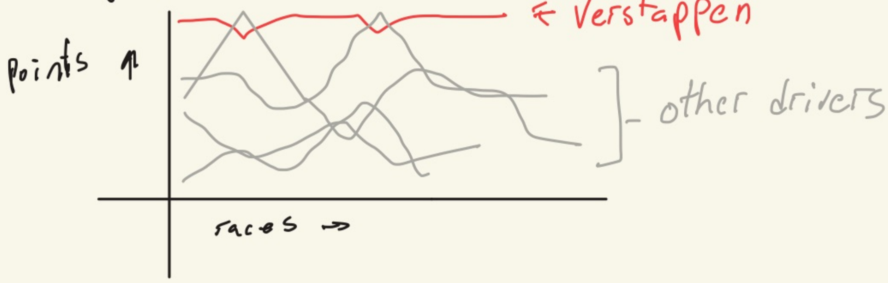
- Finalize visualization design
- Prepare presentation slides or demo
- Practice explaining insights and design decisions
- Deliver project presentation

Week 7 (May 15 – May 20) – Final Submission

Deadlines: Project Report Draft (May 17), Final Submission (May 20)

- Write and finalize project report
- Document methodology and insights
- Perform final polishing of visualization
- Submit final project

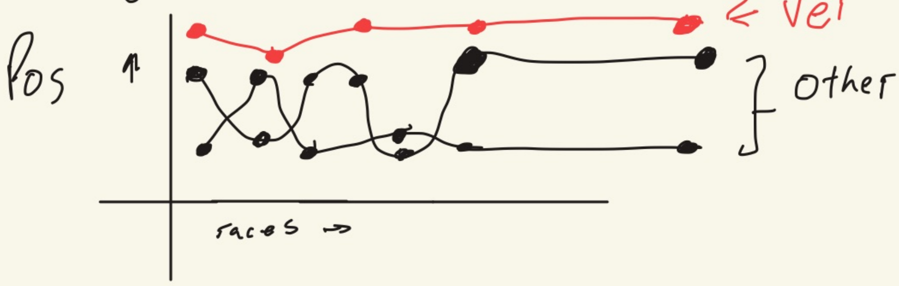
Prototype 1 → Perf Timeline



Line Chart:

→ Clearly shows performance
→ Easy to understand

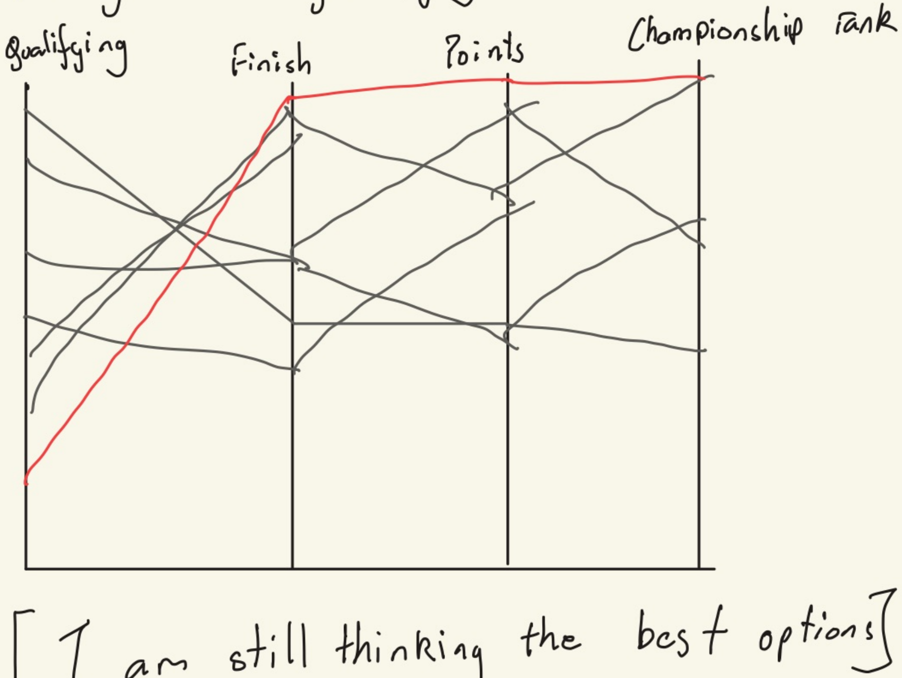
Prototype 2 → Championship Comparison



Bump Chart:

→ Clearly shows who is leading and when
→ Highights dominance.

Prototype 3: Qualifying vs Race Perf



[I am still thinking the best options]

Parallel Coordinate Chart:

→ Captures multiple components
→ Easy to notice trends
→ Room for deeper analysis

Resources used for this

Only use of generative AI was to generate a rough project schedule and brush up words.

Reference for bump chart - https://public.tableau.com/app/profile/nastengraph/viz/MaxVerstappen10Formula1Wins/Bump

This reference for information and will perhaps be used for brushing up the dataset - https://www.formula1points.com/driver/Max%20Verstappen

Other websites looked at -

https://python.plainenglish.io/analyzing-the-performance-of-formula-1-drivers-in-the-2023-season-3f8d6f305cf5

https://www.sportschord.com/en-us/blogs/blog/dont-verstappen-me-now-using-data-visualization-to-benchmark-max-verstappen-against-the-f1-greats